It from Bit: A Concrete Attempt

Emergent physics from a bit-local lattice automaton

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Resumo

John Archibald Wheeler's dictum "it from bit" holds that every physical object—every *it*—derives its existence, its name, and its laws from bits, from information-theoretic answers to yes-or-no questions [1, 2]. This manuscript presents a concrete, fully executable attempt at that program: a *toy universe* in which geometry, a wave-like light clock, charge, polarization, and a matter–antimatter asymmetry all emerge as degrees of freedom of a single deterministic lattice automaton. The model is a 3-torus cellular automaton whose cells carry six charge bits updated by integer arithmetic only, organized into winding layers with an island-affinity structure. A pulsating spherical wavefront—a constant-speed "light clock"—schedules all local dynamics; an election–broadcast–reconstruction protocol lets a transverse polarization pair (u, v) emerge from pure bit propagation; and a capture/escape balance at the level of islands realizes a dynamic charge-quantization attractor. The automaton is instrumented end to end: shell completeness, radial amplitude profiles, charge census, and island-population statistics are sampled every light frame. We argue that the construction is a faithful, if miniature, instance of the *it from bit* thesis, and we discuss which features are model-dependent engineering and which hint at genuinely information-theoretic origins.

Resumo

O ditado de John Archibald Wheeler, *it from bit* ("o isto vem do bit"), sustenta que todo objeto físico—todo "isto"—deriva sua existência, seu nome e suas leis de bits, das respostas informacionais a perguntas de sim ou não. Este manuscrito apresenta uma tentativa concreta e plenamente executável desse programa: um universo de brinquedo no qual a geometria, um relógio de luz ondulatório, a carga, a polarização e uma assimetria matéria–antimatéria emergem como graus de liberdade de um único autômato celular determinístico. O modelo é um autômato celular sobre um 3-toro cujas células carregam seis bits de carga, atualizadas apenas com aritmética inteira, organizadas em camadas de enrolamento com uma estrutura de afinidade por ilhas. Uma frente de onda esférica pulsante—um "relógio de luz" de velocidade constante—agenda toda a dinâmica local; um protocolo de eleição–difusão–reconstrução faz emergir um par de polarização transversal (u, v) a partir da pura propagação de bits; e um balanço captura/escape ao nível das ilhas realiza um atrator de quantização dinâmica de carga. O autômato é instrumentado de ponta a ponta: completude de casca, perfis radiais de amplitude, censo de carga e estatísticas de população de ilhas são amostrados a cada quadro de luz. Argumentamos que a construção é uma instância fiel, ainda que miniatura, da tese do *it from bit*, e discutimos quais características são engenharia dependente do modelo e quais sugerem origens genuinamente informacionais.

Keywords: it from bit, digital physics, cellular automata, emergent spacetime, emergent charge, wavefront automaton, computational universe.

*Source code of the *Toy Universe* automaton: the `alpha` repository.

1 Introduction

"It from bit—otherwise put, every *it*—every particle, every field of force, even the spacetime continuum itself—derives its function, its meaning, its very existence ... from the answers to yes-or-no questions ... from bits" [1, 2]. With these words John Archibald Wheeler posed the strongest form of the thesis that physics is, at bottom, information processing. The slogan has since become the banner of *digital physics*: the conjecture that the universe is a discrete, programmable system—a computer of sorts—and that its apparent continuity is an emergent fiction [3, 4, 5, 6, 7].

The program is easy to state and notoriously hard to carry out. Most proposals remain at the level of metaphor: a slogan, a research programme, a dictionary between information-theoretic and physical vocabulary, but no executable object whose dynamics actually *produce* particles, charges, or geometry. This manuscript is a deliberate, engineering-flavored counterpoint. We present a complete, running, fully instrumented automaton—a *toy universe*—in which the central items of the physics vocabulary are *read off* a single bit-local update rule rather than postulated as fields.

The construction follows five design choices, each intended to be as concrete as possible.

1. **Everything is a bit.** The fundamental state is a lattice of cells, each carrying six charge bits and a handful of auxiliary flags. There are no real-valued fields: every update is an integer operation.

2. **Geometry is computed, not stored.** Squared distances and integer radii from each source center are maintained by an incremental, add-only integer algorithm, so a spherical shell is an emergent set of cells rather than a stored mathematical surface.

3. **Time is a local light clock.** A pulsating wavefront—an expanding and contracting spherical shell of period $2R_{\max}$— replaces the continuous time parameter of field theory with a discrete, locally available signal.

4. **Structure emerges through broadcast.** The transverse polarization pair (u, v) and the momentum axis are not stored state: every cell *reconstructs* them from a single scalar "arrival stamp" broadcast by a helical walker.

5. **Everything is measured.** The automaton samples its own observables—shell completeness, radial amplitude profiles, charge census, island populations—every light frame, yielding quantitative output comparable against the expectations of a constant-speed wave.

The remainder of this paper is organized as follows. Section 2 defines the automaton. Section 3 describes the built-in instrumentation. Section 4 summarizes the implementation. Section 5 reports representative observations. Section 6 discusses, in Wheeler's spirit, what the toy universe can and cannot teach us about "it from bit" as a physical thesis. Section 7 concludes.

2 The model

2.1 Lattice, cell state, and update schedule

The arena is a three-torus of edge length L: coordinates $x, y, z \in \{0, \ldots, L-1\}$ wrap around, so a step across a face of the cube continues from the opposite face as a pure translation—orientation is preserved, nothing is mirrored. The automaton in fact allocates a stack of W such lattices, indexed by a fourth coordinate w, the *winding* index. Cross-layer adjacency is governed not by spatial geometry but by a rotation schedule (a shift-mirror operation), so the w axis is a discrete analogue of an internal (winding) degree of freedom.

Each cell of the combined lattice of size $L^3 \times W$ is a compact record:

- six *charge bits*—three colour bits c_0, c_1, c_2, one bit q, and two weak bits w_0, w_1—packed into a single byte, with masks $C_0 = 0 \times 01$, $C_1 = 0 \times 02$, $C_2 = 0 \times 04$, $Q = 0 \times 08$, $W_0 = 0 \times 10$, $W_1 = 0 \times 20$;
- an intrinsic identity w and an auxiliary *leader* identity shared by an entire winding island;
- an *affinity* label a fixing island membership, with a special value marking orphan cells;
- integer squared radius r^2 and integer radius r from the layer's moving source center;
- the phase flags p_B and s_B (local wave sign and emergent transverse flag);
- the emergent polarization pair (u, v) and the broadcast arrival stamp b.

Every tick each cell is updated from its neighbourhood by the composition of elementary operations—convolution (local averaging of the phase signal), diffusion (spreading of affinity), and relocation (reissue of released charges)—optionally with programmable delays between the phases. All operations are bit and integer operations; floating point never appears in the update kernel.

2.2 The pulsating wavefront: a lattice light clock

A single master schedule drives the entire automaton. Each winding layer owns a source center; the active shell radius is the triangle wave

$$\rho(t) \;=\; \begin{cases} \tau, & \tau \le R_{\max}, \\ 2R_{\max} - \tau, & \tau > R_{\max}, \end{cases} \qquad \tau = t \bmod 2R_{\max}, \tag{1}$$

with period $2R_{\max}$ and amplitude $R_{\max} = L/2$. A cell is *active* exactly when its propagated integer radius r equals $\rho(t)$; because the peak advances one cell per tick, the front is a constant-speed wave—a lattice-native "light clock" that, unlike a global integer time counter, is available locally to every cell. At the turnaround $t \equiv R_{\max}$ the shell stops, inverts, and *elects* a direction: this tick is the dynamical event on which the emergent structure of Section 2.4 hangs.

The radius field itself is never recomputed by square roots. It is maintained by an incremental CaRaSh-style distance transform: after each step of the source center, every cell updates r^2 and r using additions and simple square-boundary tests, so the shell "just happens" as the collection of cells whose radius matches the schedule (1).

2.3 Winding layers and island affinity

Winding layers are grouped into *islands* of fixed size $3L^2$; the chief layer of an island is its first member. Initial conditions place, in each layer, a sphere of radius $R_{\max}$ around the layer center; cells inside the sphere inherit the island chief as affinity, cells outside are orphans. Affinity is dynamic: across light frames, *capture* events turn orphans into members, *escape* events release members back into the orphan pool, and a member changing island counts as escape plus capture. These events are the fuel of the charge-quantization attractor of Section 3.

2.4 Emergent polarization pair (u, v)

Perhaps the most distinctive mechanism is the way the model refuses to store a polarization field. Instead, a three-stage protocol reconstructs it from a scalar bit signal.

Election. At every expansion limit of the breathing wavefront—the exact tick at which the wave turns from ascending to descending, i.e. $t \equiv R_{\max}$—every active shell cell is seeded with the payload

$$\text{payload}(x) = \big(\text{score}(x) \ll 24\big) \mid \text{code}(x), \qquad \text{score}(x) = H\big((\nu(x) + s) \bmod 9L,\ \text{code}(x)\big), \tag{2}$$

where $\nu(x)$ is the island index of the cell, $\text{code}(x)$ its linear index, H a fixed integer hash and s a global seed dial. Because the code occupies the low bits, the payload order is a strict total order: ties are impossible, so the winner is unique and is identified once, at sowing time. The displacement from the lattice center to the winning cell becomes the momentum vector $\mathbf{m}$, rescaled to $|\mathbf{m}| = L/2$.

Broadcast. A helical walker traces a spiral around the cylinder whose axis is the elected direction, advancing one cell per tick using only additions, subtractions, shifts, and comparisons among its six face neighbours. Each freshly computed spiral point stamps the current tick into the cell's arrival field b of a second value lattice (zero meaning "never reached"). Every other tick, with alternating sweep directions, each cell copies the greatest neighbour stamp whenever it exceeds its own, so the news diffuses to every cell in finite time.

Reconstruction. The arrival stamp is a phase source:

$$\phi_{\text{full}} = 2R^2, \qquad R = R_{\max} - 2, \qquad \phi = (b-1) \bmod \phi_{\text{full}}, \qquad j = \lfloor \phi / R \rfloor, \tag{3}$$

and the transverse pair is

$$u = \begin{cases} R(R-2j), & j < R, \\ R(2j-3R), & j \geq R, \end{cases} \qquad v = \begin{cases} 2R\,\text{isqrt}\big(j(R-j)\big), & j < R, \\ -2R\,\text{isqrt}\big((j-R)(2R-j)\big), & j \geq R, \end{cases} \tag{4}$$

where isqrt is the integer square root. The pair (4) approximates the circle $u^2 + v^2 = R^4$, exactly when the integer square root is exact. Polarization is thus not a field that evolves; it is a bit that *travels*, and a rule that *decodes*.

2.5 Charge dynamics and matter–antimatter census

The charge byte is the only particle-like memory of a cell. Its six bits are recombined by the local update rule as islands interact; the rule is written so that the q bit and the weak bits participate in the capture/escape balance that re-emits charge at each island turnaround ("clock reset"). A separate, read-only instrumentation layer maintains a per-tick census of matter and antimatter populations, split by the two global sectors (*Orbis* and *Umbra*), together with a ledger of "virgin wraps"—cells whose winding parity has never flipped. The census is purely observational: by construction it never writes to the lattice, so the automaton's dynamics are bit-identical with or without it.

3 Instrumentation and observables

A toy universe is only scientific if it can be measured. The automaton ships with four instrument modules, all reading *only* the committed state of the lattice and all emitting machine-readable reports.

Wavefront fidelity. Once per completed light frame, for each integer radius r that was the active target of some layer, the observed number of active cells is divided by the theoretical number of lattice points with $r^2 \leq \Delta x^2 + \Delta y^2 + \Delta z^2 < (r+1)^2$ (exact integer counting, no square roots). A faithful constant-speed front should reach completeness ≈ 1. The module also records the radial amplitude profile $\langle |u| \rangle(r)$ correlated against $\sin(r)/r$, and the peak-radius error—the distance between the observed peak of $\langle |u| \rangle$ and the target radius—a direct check that the wave peak advances one cell per tick.

Charge census. The matter/antimatter census of Section 2.5, throttled per tick, with capture and escape rates per island.

Dynamic charge-quantization attractor. For every island, the population $N(t)$ is sampled each light frame, together with gross capture and escape counts. The hypothesis, stated in the model as a target to be tested, is that there is a stable attractor N^*:

$$\Gamma_{\text{cap}}(N) > \Gamma_{\text{esc}}(N) \text{ for small } N, \qquad \Gamma_{\text{cap}}(N) < \Gamma_{\text{esc}}(N) \text{ for large } N, \tag{5}$$

with $\Gamma_{\text{cap}}(N^*) \approx \Gamma_{\text{esc}}(N^*)$. The module performs an OLS fit of $\Delta N(t) = N(t+1) - N(t)$ against $N(t)$; a negative slope is a restoring coefficient, and N^* is estimated as $-$intercept/slope, with standard errors. Time series can be saved to and restored from disk, and written as CSV for external analysis.

Tomography. A visualization instrument slices, inverts, and animates the volume along any axis with adjustable thickness, so that interior structure—shells, elected axes, island shapes—can be inspected directly.

4 Implementation

The automaton is written in C++17. The simulation runs on a dedicated thread; rendering and interaction run on the main thread, and the voxel buffer is guarded by a mutex so the two never race. A condition variable synchronizes thread readiness; all cross-thread flags are atomic. The renderer uses OpenGL 4.6 core profile through GLFW, GLAD and GLM, with a full HUD, menus, recorder/replay pipeline, and a 2D/3D projection manager. An optional CUDA bridge mirrors the three lattices on the device and accelerates the update kernel when a GPU is available; a host-side stub preserves identical semantics otherwise. All run parameters—lattice size, scenario, delays, field visualization toggles, projection, tomography settings—are read from a plain-text configuration file at startup, so every reported number is reproducible.

5 Results

The instrumentation is designed so that three claims can be checked quantitatively at runtime.

The front is a faithful light clock. Shell completeness near unity and a peak-radius error of at most one cell per light frame confirm that the integer distance transform and the triangle schedule (1) really implement a constant-speed front, and that the "speed of light" is a property of the rule, not of a stored time parameter.

Polarization is a travelling bit. The reconstruction (3)–(4) predicts that every cell that ever receives a nonzero arrival stamp lies, up to integer-square-root error, on the circle $u^2 + v^2 = R^4$; monitoring the distribution of $u^2 + v^2 - R^4$ over the lattice converts "emergent polarization" into a testable bound on the automaton.

Charge quantizes. A negative fitted slope in (5) with a well-constrained N^* demonstrates that the capture/escape balance self-organizes a preferred charge population per island—a genuinely emergent conservation law, in the sense that no bit of the rule states it explicitly.

A full quantitative report, with per-run tables and figure panels produced by the headless runner, is the subject of a companion document; this draft concentrates on the architecture and on the falsifiable statements the instrumentation exists to test. We stress what the toy universe does *not* claim: it does not quantize a pre-existing field, it does not store a metric, and it does not invoke randomness—it is deterministic and scale-free. Its virtue is that every emergent word in the previous paragraphs has a precise, executable meaning.

6 Discussion

Wheeler asked whether physics might be, at bottom, a matter of "bits answering yes-or-no questions." The toy universe is a concrete laboratory for that question, and its architecture exposes three lessons.

First, *emergence without a substrate.* Charge, momentum, and polarization here are not fields that live on a pre-existing geometry; they are decoded from propagation histories on a bit lattice whose geometry is itself maintained by an integer transform. There is no level at which one is forced to introduce a real number.

Second, *the price of emergence is instrumentation.* A polarization pair that is reconstructed rather than stored is indistinguishable from noise unless the automaton measures it. Every term in the model's emergent vocabulary therefore comes paired with an observable—shell completeness, $u^2 + v^2 - R^4$, the fitted slope of ΔN versus N. This pairing is, we believe, the methodological core of any serious "it from bit" programme.

Third, *the honest objection.* Why this rule? A generic lattice automaton does not look like a sphere, a polarization pair, or an attractor; the design choices here are precisely the ones that produce those features. This objection is not an argument against "it from bit"—it *is* the research programme the slogan names. If physics is computation, the physical laws are a program, and programs have authors; the question "which program, and why" is exactly the question Wheeler bequeathed to us.

7 Conclusion

We have presented a complete, running automaton—a toy universe—in which a spherical light clock, a momentum axis, a transverse polarization pair, island-affine charge, and a matter–antimatter census all derive from a single bit-local rule, without any of them being written into the initial conditions as continuous fields. The construction demonstrates, at least in miniature, the two theses that make "it from bit" more than a slogan: that the standard-model vocabulary *can* be instantiated by a purely combinatorial system, and that emergence must be measured to be believed. The instrumentation is the arbiter of the second thesis, and the choice of rule is the open frontier of the first.

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